Ilia Mirzaali

Software Engineer

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Profile

Software engineer and second-year Computer Science student at the University of Twente (BSc, expected Jul. 2028; weighted average 8.42/10). Build and operate web products, develop C++ and Linux systems tools, and run Ramsy, a security consultancy. Experience spans customer requirements, application design, automated testing, cloud deployment and production reliability.

Experience

Ramsy (ramsy.eu) Enschede, Netherlands
Founder & Engineer *Mar. 2026 – Present*

- Founded and run a KvK-registered security consultancy serving Dutch businesses; delivered 20+ engagements from scoping and pricing through testing, reporting and retesting.
- Built a React and TypeScript client portal on Supabase/PostgreSQL, with organisation-scoped access policies, private evidence storage, and tracking of findings, remediation and retest status.
- Developed a publishing pipeline that validates client-facing findings, excludes internal assessment data, and keeps portal content consistent with delivered reports.
- Built internal tooling for asset and subdomain discovery, scan orchestration and report generation; explain technical findings and remediation priorities to customers.

Quicksports B.V. Enschede, Netherlands
Software Engineer (startup contributor) *Oct. 2025 – Present*

- Own the Organizer Dashboard from customer requirements to release, building full-stack features in Vue 3, Vuetify and Node.js and iterating directly on customer feedback.
- Lead and mentor developers and interns through onboarding, task planning and code review.
- Built the real-time messaging service on Node.js, Socket.IO and Redis Pub/Sub, with MongoDB-backed caching.
- Reworked scheduled jobs and notification delivery with durable MongoDB records, retryable processing and dedicated workers; verified delivery, restart recovery and production monitoring.
- Migrated 5 microservices from Heroku to Google Cloud Run with Cloud Build CI/CD; tuned container deployments and autoscaling, reducing infrastructure spend by 30%.

Full Stack Developer (startup contributor) *May 2025 – Oct. 2025*

- Shipped the Communities feature and rebuilt the admin dashboard during a period of 20% user-base growth.
- Built REST APIs and internal tools on Node.js, Express, MongoDB and Redis, including testing, CI/CD and deployment.

DroneTeam Twente Enschede, Netherlands
Chief of DronoBox *Apr. 2026 – Present*

- Built and maintain droneteamtwente.nl, the team's public website.

DronoBox Engineer *Sep. 2025 – Apr. 2026*

- Co-organised drone workshops, mentored students in building and flying drones, and maintained and repaired the team's drone fleet.

Achievements & Activities

- **1st place of 13 teams, Amsterdam Fixathon 2026** (Stichting Single SuperMom challenge): co-built an AI-assisted Dutch benefits eligibility engine using hybrid search and machine-checkable rules.
- **BAPC 2025**: 2nd among University of Twente teams; qualified for the Northwestern European Regional Contest (NWERC).
- **3rd place, Twente Algorithmic Programming Contest (TAPC) 2025.**
- **Co-founder, C3ntaur**, a student CTF team; 48th internationally at Nullcon Goa HackIM 2026.

Selected Projects

RayaChat – AI Customer-Support Platform *TypeScript, Next.js, PostgreSQL*

- Developed and deployed a customer-support platform with web and Telegram channels, an embeddable chat widget, conversation inbox and usage-based billing.
- Implemented a document-to-answer pipeline for uploaded files and URL sources, using embeddings, PostgreSQL/pgvector and retrieval-augmented generation.
- Built a developer API with API keys, idempotent requests and signed webhooks; added integration tests and Docker/GitHub Actions deployment with backup and health checks.

Khor – Kernel Activity into Music *C++, eBPF, React, TypeScript*

- Built a Linux daemon that turns live process, network, scheduler and I/O metrics into music using eBPF collection and a deterministic mapping engine.
- Implemented a custom C++ polyphonic synthesiser with MIDI and OSC outputs, plus a React/TypeScript dashboard for live control and telemetry.

ChokerJoker – Quarto Game Engine *C++, Java, Search Algorithms*

- Won the University of Twente Module 2 algorithms competition with a game-playing engine combining heuristic midgame search and endgame solving.
- Extended the Java solver to C++ with parallel search, bitboards, Zobrist hashing, symmetry reduction and a simulation harness for comparing agents.

Cedis – Redis-Protocol Server *C++17, Boost.Asio, CMake*

- Designed an asynchronous TCP key-value server implementing a RESP2 subset, six Redis commands and a shared in-memory store.
- Built a modular parser and command dispatcher, with CMake builds and GitHub Actions CI for Linux, macOS and Windows.

Other work: CHIP-8 interpreter (C++20, SDL2, 35 standard opcodes); ilia.beer (Next.js/PHP, payments, Raspberry Pi hosting and Google Cloud video transcoding). Write technical articles on systems projects, algorithms and security research at rmxzy.com/blog.

Education

University of Twente Enschede, Netherlands
BSc Computer Science – expected Jul. 2028 *Sep. 2025 – Present*
Weighted average: 8.42/10.

Twente Pathway College Enschede, Netherlands
Foundation Year, General Engineering *Sep. 2024 – Jun. 2025*

NODET (Sampad) Gorgan, Iran
High School Diploma, Mathematics & Physics *Graduated Jun. 2023*
Admission by nationwide competitive examination.

Skills

Programming: C, C++, TypeScript, JavaScript, Python, Java, PHP, SQL, Bash; Rust (learning)

Web & APIs: React, Next.js, Vue 3, Vuetify, Node.js, Express, Hono, FastAPI, Flask, REST APIs, WebSockets

Data & AI: PostgreSQL, pgvector, MongoDB, Redis, Supabase, Prisma; document ingestion and retrieval

Systems: Linux, eBPF, Boost.Asio, CMake, SDL2, systemd; asynchronous I/O and concurrency

Cloud & delivery: Docker, Google Cloud (Run, Compute Engine, Cloud Build), AWS (EC2, S3, Lambda), Nginx, GitHub Actions, GitLab CI, Git

Testing & operations: Vitest, unit and integration tests, production monitoring, Netdata, Uptime Kuma

Security: Web and API testing, secure code review, OWASP Top 10/WSTG, Burp Suite, threat modelling

Languages: Persian and Turkmen (native), English (C1), Turkish (B2), Dutch (A1, actively learning)